

4 The Fourier Transform

Raymond Edward Alan Christopher Paley, Fellow of Trinity College, Cambridge, and International Research Fellow at the Massachusetts Institute of Technology and at Harvard University, was killed by an avalanche on April 7, 1933, while skiing in the vicinity of Banff, Alberta. Although only twenty-six years of age, he was already recognized as the ablest of the group of young English mathematicians who have been inspired by the genius of G. H. Hardy and J. E. Littlewood. In a group notable for its brilliant technique, no one had developed this technique to a higher degree than Paley. Nevertheless he should not be thought of primarily as a technician, for with this ability he combined creative power of the first order. As he himself was wont to say, technique without “rugger tactics” will not get one far, and these rugger tactics he practiced to a degree that was characteristic of his forthright and vigorous nature.

Possessed of an extraordinary capacity for making friends and for scientific collaboration, Paley believed that the inspiration of continual interchange of ideas stimulates each collaborator to accomplish more than he would alone. Only the exceptional man works well with a partner, but Paley had collaborated successfully with many, including Littlewood, Pólya, Zygmund, and Wiener.

N. Wiener, 1933

If f is a function on $\mathbb{R}$ that satisfies appropriate regularity and decay conditions, then its Fourier transform is defined by

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi} dx, \quad \xi \in \mathbb{R}$$

and its counterpart, the Fourier inversion formula, holds

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x \xi} d\xi, \quad x \in \mathbb{R}.$$

The Fourier transform (including its d -dimensional variants), plays a ba-

sic role in analysis, as the reader of Book I is aware. Here we want to illustrate the intimate and fruitful connection between the one-dimensional theory of the Fourier transform and complex analysis. The main theme (stated somewhat imprecisely) is as follows: for a function f initially defined on the real line, the possibility of extending it to a holomorphic function is closely related to the very rapid (for example, exponential) decay at infinity of its Fourier transform $\hat{f}$. We elaborate on this theme in two stages.

First, we assume that f can be analytically continued in a horizontal strip containing the real axis, and has “moderate decrease” at infinity,¹ so that the integral defining the Fourier transform $\hat{f}$ converges. As a result, we conclude that $\hat{f}$ decreases exponentially at infinity; it also follows directly that the Fourier inversion formula holds. Moreover one can easily obtain from these considerations the Poisson summation formula $\sum_{n \in \mathbb{Z}} f(n) = \sum_{n \in \mathbb{Z}} \hat{f}(n)$. Incidentally, all these theorems are elegant consequences of contour integration.

At a second stage, we take as our starting point the validity of the Fourier inversion formula, which holds if we assume that both f and $\hat{f}$ are of moderate decrease, without making any assumptions on the analyticity of f . We then ask a simple but natural question: What are the conditions on f so that its Fourier transform is supported in a bounded interval, say $[-M, M]$? This is a basic problem that, as one notices, can be stated without any reference to notions of complex analysis. However, it can be resolved only in terms of the holomorphic properties of the function f . The condition, given by the Paley-Wiener theorem, is that there be a holomorphic extension of f to $\mathbb{C}$ that satisfies the growth condition

$$|f(z)| \leq Ae^{2\pi M|z|} \quad \text{for some constant } A > 0.$$

Functions satisfying this condition are said to be of **exponential type**.

Observe that the condition that $\hat{f}$ vanishes outside a compact set can be viewed as an extreme version of a decay property at infinity, and so the above theorem clearly falls within the context of the theme indicated above.

In all these matters a decisive technique will consist in shifting the contour of integration, that is the real line, within the boundaries of a horizontal strip. This will take advantage of the special behavior of $e^{-2\pi iz\xi}$ when z has a non-zero imaginary part. Indeed, when z is real this exponential remains bounded and oscillates, while if $\text{Im}(z) \neq 0$, it will

¹We say that a function f is of **moderate decrease** if f is continuous and there exists $A > 0$ so that $|f(x)| \leq A/(1+x^2)$ for all $x \in \mathbb{R}$. A more restrictive condition is that $f \in \mathcal{S}$, the Schwartz space of testing functions, which also implies that $\hat{f}$ belongs to $\mathcal{S}$. See Book I for more details.

have exponential decay or exponential increase, depending on whether the product $\xi \operatorname{Im}(z)$ is negative or positive.

1 The class $\mathfrak{F}$

The weakest decay condition imposed on functions in our study of the Fourier transform in Book I was that of moderate decrease. There, we proved the Fourier inversion and Poisson summation formulas under the hypothesis that f and $\hat{f}$ satisfy

$$|f(x)| \leq \frac{A}{1+x^2} \quad \text{and} \quad |\hat{f}(\xi)| \leq \frac{A'}{1+\xi^2}$$

for some positive constants A, A' and all $x, \xi \in \mathbb{R}$. We were led to consider this class of functions because of various examples such as the Poisson kernel

$$P_y(x) = \frac{1}{\pi} \frac{y}{y^2 + x^2}$$

for $y > 0$, which played a fundamental role in the solution of the Dirichlet problem for the steady-state heat equation in the upper half-plane. There we had $\widehat{P}_y(\xi) = e^{-2\pi y|\xi|}$.

In the present context, we introduce a class of functions particularly suited to the goal we have set: proving theorems about the Fourier transform using complex analysis. Moreover, this class will be large enough to contain many of the important applications we have in mind.

For each $a > 0$ we denote by $\mathfrak{F}_a$ the class of all functions f that satisfy the following two conditions:

- (i) The function f is holomorphic in the horizontal strip

$$S_a = \{z \in \mathbb{C} : |\operatorname{Im}(z)| < a\}.$$

- (ii) There exists a constant $A > 0$ such that

$$|f(x + iy)| \leq \frac{A}{1+x^2} \quad \text{for all } x \in \mathbb{R} \text{ and } |y| < a.$$

In other words, $\mathfrak{F}_a$ consists of those holomorphic functions on S_a that are of moderate decay on each horizontal line $\operatorname{Im}(z) = y$, uniformly in $-a < y < a$. For example, $f(z) = e^{-\pi z^2}$ belongs to $\mathfrak{F}_a$ for all a . Also, the function

$$f(z) = \frac{1}{\pi} \frac{c}{c^2 + z^2},$$

which has simple poles at $z = \pm ci$, belongs to $\mathfrak{F}_a$ for all $0 < a < c$.

Another example is provided by $f(z) = 1/\cosh \pi z$, which belongs to $\mathfrak{F}_a$ whenever $|a| < 1/2$. This function, as well as one of its fundamental properties, was already discussed in Example 3, Section 2.1 of Chapter 3.

Note also that a simple application of the Cauchy integral formulas shows that if $f \in \mathfrak{F}_a$, then for every n , the n^{th} derivative of f belongs to $\mathfrak{F}_b$ for all b with $0 < b < a$ (Exercise 2).

Finally, we denote by $\mathfrak{F}$ the class of all functions that belong to $\mathfrak{F}_a$ for some a .

Remark. The condition of moderate decrease can be weakened somewhat by replacing the order of decrease of $A/(1+x^2)$ by $A/(1+|x|^{1+\epsilon})$ for any $\epsilon > 0$. As the reader will observe, many of the results below remain unchanged with this less restrictive condition.

2 Action of the Fourier transform on $\mathfrak{F}$

Here we prove three theorems, including the Fourier inversion and Poisson summation formulas, for functions in $\mathfrak{F}$. The idea behind all three proofs is the same: contour integration. Thus the approach used will be different from that of the corresponding results in Book I.

Theorem 2.1 *If f belongs to the class $\mathfrak{F}_a$ for some $a > 0$, then $|\hat{f}(\xi)| \leq Be^{-2\pi b|\xi|}$ for any $0 \leq b < a$.*

Proof. Recall that $\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x)e^{-2\pi i x \xi} dx$. The case $b = 0$ simply says that $\hat{f}$ is bounded, which follows at once from the integral defining $\hat{f}$, the assumption that f is of moderate decrease, and the fact that the exponential is bounded by 1.

Now suppose $0 < b < a$ and assume first that $\xi > 0$. The main step consists of shifting the contour of integration, that is the real line, down by b . More precisely, consider the contour in Figure 1 as well as the function $g(z) = f(z)e^{-2\pi i z \xi}$.

We claim that as R tends to infinity, the integrals of g over the two vertical sides converge to zero. For example, the integral over the vertical segment on the left can be estimated by

$$\begin{aligned} \left| \int_{-R-ib}^{-R} g(z) dz \right| &\leq \int_0^b |f(-R-it)e^{-2\pi i(-R-it)\xi}| dt \\ &\leq \int_0^b \frac{A}{R^2} e^{-2\pi t \xi} dt \\ &= O(1/R^2). \end{aligned}$$